

Periodic Motion and Oscillations:

A motion which repeats along the same path after a definite interval of time, is called periodic motion.

Earth's motion round the sun, motion of the hands (hour, minute) of a clock, motion of planets etc are the examples of periodic motion. The rotation of the earth about the sun is periodic and its period 365 days.

When a body moves back and forth repeatedly about a mean position, its motion is called Oscillation or Vibration.

Alternatively, if the periodic motion of an object is such that during half of its time period it stays in one direction and during rest half in the opposite direction then that type of motion is called Oscillation or Vibration.

Motion of a simple pendulum, vibration of tuning fork, motion of a spring etc are the examples of Oscillation or Vibration.

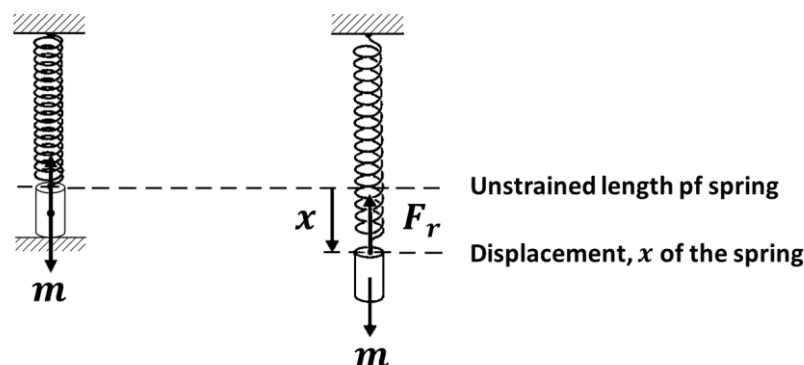
We can say that “all oscillations or vibrations are periodic motion, but all periodic motions are not oscillations or vibrations”

Simple Harmonic Oscillation:

The type of vibratory motion of a body such that the restoring force or the acceleration acting on the body is directly proportional to the displacement from the mean position and always directed towards the mean position is called the Simple Harmonic Oscillation (SHO) or Motion.

The direct proportionality of acceleration and displacement distinguishes simple harmonic motion from all type of vibratory motion. This motion is always motion along a straight line.

Explanation: Suppose an object of mass, m , hanging at the end of a spiral spring is pulled down a distances below the equilibrium position.



The spring exerts a restoring force on the object, tending to pull it back towards its equilibrium position. This force is proportional to the displacement, x but opposite in direction i.e.,

$$F \propto -x$$

$$F = -Kx$$

Where, K is a constant, called Force constant or spring constant. When the object is released, the restoring force produces an acceleration. This acceleration is proportional to the displacement but opposite in direction.

$$ma = -Kx$$

$$a \propto -x$$

Characteristics of Simple Harmonic Oscillation:

A simple harmonic oscillation has the following characteristics:

1. Its motion is periodic.
2. At particular time interval the motion becomes opposite.
3. Its motion is along a straight line.
4. Its acceleration is proportional to the displacement.
5. Acceleration is opposite to displacement.
6. Acceleration points toward the mean position of the object.

Differential Equation of SHO:

Let a particle of mass oscillate in simple harmonic motion. Now, at time t if its displacement is x , then –

$$\text{Velocity, } v = \frac{dx}{dt} \quad \text{and acceleration, } a = \frac{d^2x}{dt^2}$$

$\therefore$ Magnitude of force acting on the particle,

$$F = ma = m \frac{d^2x}{dt^2}$$

Now,

$$m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Now, we defined as, the angular velocity, $\omega^2 = \frac{k}{m}$;

$$\frac{d^2x}{dt^2} + \omega^2x = 0$$

This is the differential equation of a particle executing simple harmonic oscillation.

Solution of the above equation:

By solving the above equation, we get the displacement of the particle.

In order to solve, let us multiply both sides by $\frac{2dx}{dt}$ then,

$$\therefore 2 \frac{dx}{dt} \times \frac{d^2x}{dt^2} + \omega^2x \times 2 \frac{dx}{dt} = 0$$

Integrating the above equation, we get –

$$\left(\frac{dx}{dt}\right)^2 + \omega^2x^2 = c$$

Here, c is the integration constant. Now, when $x = A$, then velocity, $\frac{dx}{dt} = 0$

Inserting this condition, we get –

$$c = \omega^2A^2$$

$$\therefore \left(\frac{dx}{dt}\right)^2 = \omega^2(A^2 - x^2)$$

$$\Leftrightarrow \frac{dx}{dt} = \sqrt{\omega^2(A^2 - x^2)} = \omega\sqrt{(A^2 - x^2)}$$

$$\Leftrightarrow \frac{dx}{\sqrt{(A^2 - x^2)}} = \omega dt$$

Integrating the above equation, we get –

$$\sin^{-1}\left(\frac{x}{A}\right) = \omega t + \delta$$

Here, δ is the integration constant.

$$x = A\sin(\omega t + \delta)$$

Some terms relating to SHM:

(a) Time Period, Frequency & Angular frequency: Time taken for one complete oscillation of the particle executing simple harmonic motion is called time period.

$$T = \frac{t}{N}$$

Also, The total number of oscillations completed by an oscillating particle in one second is called its frequency.

$$f = \frac{N}{t}$$

So, relation between time period and frequency is –

$$T = \frac{1}{f}$$

Now, the angular distance travelled by a particle executing simple harmonic motion is called angular frequency.

$$\omega = \frac{2\pi}{T}$$

Again, $\omega = \sqrt{\frac{K}{m}}$; so – $\sqrt{\frac{K}{m}} = \frac{2\pi}{T}$

$$\therefore T = 2\pi \sqrt{\frac{m}{K}}$$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{K}{m}}$$

(b) Velocity: The displacement of a particle executing simple harmonic motion is given by

$$x = A \sin(\omega t + \delta)$$

Differentiating this displacement with respect to time (t), then we get the velocity,

$$v = \frac{dx}{dt} = \omega A \cos(\omega t + \delta) = \omega A \sqrt{1 - \sin^2(\omega t + \delta)}$$

$$\Leftrightarrow v = \omega \sqrt{A^2 - A^2 \sin^2(\omega t + \delta)}$$

$$v = \omega \sqrt{A^2 - x^2}$$

(c) Acceleration: We know, the rate of change of velocity is called acceleration, i.e.,

Now,

$$a = \frac{dv}{dt} = -\omega^2 A \sin(\omega t + \delta)$$

$$a = -\omega^2 x$$